MUBASHIR TANWAR

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EXPERIENCE

SDE-1 @ ***Prana (PranaIndia)*** Jul 2025 - Present, *Bengaluru*

- Built the **real-time chat and voice-call** consultation service from scratch over **WebSockets**, replacing paid third-party APIs: messages are buffered in **Redis** and batch-flushed to the database, sustaining **150+ concurrent chats at sub-200ms latency** with AES-GCM encrypted messaging.
- Built a **Node.js/TypeScript** (Express) **microservices** backend with **BullMQ** workers queuing notifications and background jobs, deployed on **AWS EC2 with auto-scaling**, ECR and Amplify using **blue-green deployments**.
- Built an in-house **A/B testing and user-behaviour tracking library** (client-side event batching, server bulk-writes), running **20+ experiments across 150,000+ weekly users** that lifted landing-to-consultation conversion **4x (2% → 8%)**.
- Built the customer-facing pranaindia.com (**Next.js**/TypeScript), the wallet system and office.pranaindia.com; worked on an **LLM/RAG** chatbot over users' birth-chart data.

Software Developer Intern @ ***ORIM Advisors*** Jan 2025 - Apr 2025, *Mumbai*

- Built **data-extraction pipelines**: scrapers feeding an **LLM question-answering step with RAG** that turned raw data into **structured output**, plus trading-data APIs and dashboards (**FastAPI**, Celery, **Postgres**).
- Built the front-end console for an internal cloud platform in **React/TypeScript**/Tailwind, using **TanStack Query** and Zustand for server- and client-state management.
- Led an in-house, **accessible component library** adopted across multiple internal services, cutting **∼2 dev-days/week** of duplicated UI work.

Deep Learning Intern @ ***National Informatics Centre*** Jan 2024 - Oct 2024, *Mumbai*

- **Document intelligence:** fine-tuned a **TrOCR Vision Encoder-Decoder** (ViT encoder + Marathi-BERT-v2 decoder) in PyTorch and Hugging Face to **83% accuracy** on handwritten Marathi government letters.
- **Automated 90%** of labelling to build a **135,000+** sample corpus, and shipped a production **Django API** for PDF OCR with **Elasticsearch**-backed deduplication, which **passed alpha testing** in the NIC Mumbai office.

Full Stack Developer @ ***FuzzyNodes*** Feb 2024 - Sep 2024, *Remote*

- Rebuilt the bestfares365 CRM and airline-booking platform on the **MERN stack** with the **Amadeus API**, replacing the legacy system across customer and travel-agent channels.
- Implemented a **fare-search algorithm** over route and connection combinations that **surfaces cheaper itineraries** than direct queries, plus an **agent wallet** and a per-route **dynamic-pricing engine**.

PROJECTS

Outpost — Self-hosted chess coach (live) — GitHub *Next.js, FastAPI, Neo4j, LangGraph*

Built **two LangGraph agents**: a day-review pipeline, and a coach that **routes each question by intent** and **streams answers token by token over SSE** using LangGraph's custom stream writer. The **engine and the graph supply every fact** (eval swings, rating changes, recurring motifs); the LLM (Claude or Ollama) only writes the wording, so it **cannot make up analysis**. Designed a **Neo4j schema that keys every position by FEN**, so the coach can find the **same mistake across games** and generate puzzles from it. Wrote the browser's **UCI client** so a single **Stockfish worker** serves every view, which is what lets the whole **game review run with the backend switched off**.

Sudoku — Solver / Generator — GitHub *C++*

Benchmarked **backtracking** against **Knuth's Dancing Links** (Algorithm X over a **324-constraint exact-cover matrix**). On a puzzle built to break backtracking, DLX visits **81 nodes vs 69 million** (**0.03ms vs 1.9s**); the generator uses DLX to **guarantee a unique solution**.

Timetable Generator (UCTP) — GitHub *Python*

Genetic algorithm for the University Course Timetabling Problem: **weighted hard and soft constraint penalties**, mutation **aimed at the events causing clashes** and an adaptive rate to escape local optima, reaching **zero conflicts for 72 weekly lectures in ∼30s**.

EDUCATION

M.H. Saboo Siddik College of Engineering (MHSSCE), University of Mumbai 2021 - 2025

Bachelor of Engineering in Computer Science Engineering (AIML), CGPA: 8.0/10.0

SKILLS

Languages:	TypeScript, JavaScript, Python, C++, Go	**Databases:**	PostgreSQL, MongoDB, Redis, Neo4j
Frontend:	React, Next.js, TanStack Query, Redux	**AI / LLM:**	RAG, LangGraph, LangChain
Backend:	Node.js, Express, BullMQ, Socket.IO, FastAPI	**DevOps:**	AWS (EC2, S3, ECR), Docker
AI tooling:	Claude Code (daily agentic coding)	**Tools:**	Git, Postman, Prisma, Vercel